Medicine and Diagnostics

The advent of Nanomedicine

One of the hottest areas of research is nanomedicine, and Nanotechnology stands a good chance of revolutionizing the practice of medicine. If scientists domesticate atoms and molecules, they could harness them for a wide range of medical purposes. For one thing, they could create novel nanostructures that serve as new kinds of drugs for treating common conditions such as cancer, Parkinson's and cardiovascular disease. They could also engineer nanomaterials for use as artificial tissues that would replace diseased kidneys and livers, and even repair nerve damage. Moreover, they could integrate nanodevices with the nervous system to create implants that restore vision and hearing and build prosthetic limbs that better serve the disabled.

Medical applications of nanotechnology—more so than energy applications—are already rapidly developing. Many companies, including pharmaceutical giants, medicaldevice makers, biotech companies and start-up ventures, are now exploring and using nanotechnology for healthcare applications. In fact the U.S. Food and Drug Administration (FDA) approved the first drug to employ nanotechnology, more than six years ago. Called Abraxane, this nanomedicine was made by loading the drug Taxol into nanoparticles of albumin, a protein that is plentiful in human serum. In this formulation the drug appears to be less toxic than regular Taxol and more effective in treating metastatic breast cancer. The first-generation nanomedicines are mostly reformulations of existing drugs, and they often use new methods of delivery within the body. Researchers are currently testing everything from buckyballs to nanocapsules to dendrimers as vehicles for efficiently ferrying drugs in the body, particularly for delivering chemotherapy agents to tumors.

The first order of business in the application of nanotechnology in the medicine sector is to come up with faster, cheaper diagnostic equipment. For instance, a diagnostic kit that employs gold nanoparticles is being tested in hospitals for use in detecting prostate cancer early, at a stage when the numbers of protein biomarkers in the blood are quite low. The lab-on-a-chip will be able to analyze a patient's ailments in an instant, providing point-of-care testing and drug application. New contrast agents will float through the bloodstream, lighting up problems such as tumors with incredible accuracy. Diagnostic tests will become more portable, providing time-sensitive diagnostics out in the field on ambulances.


Lab on a Chip